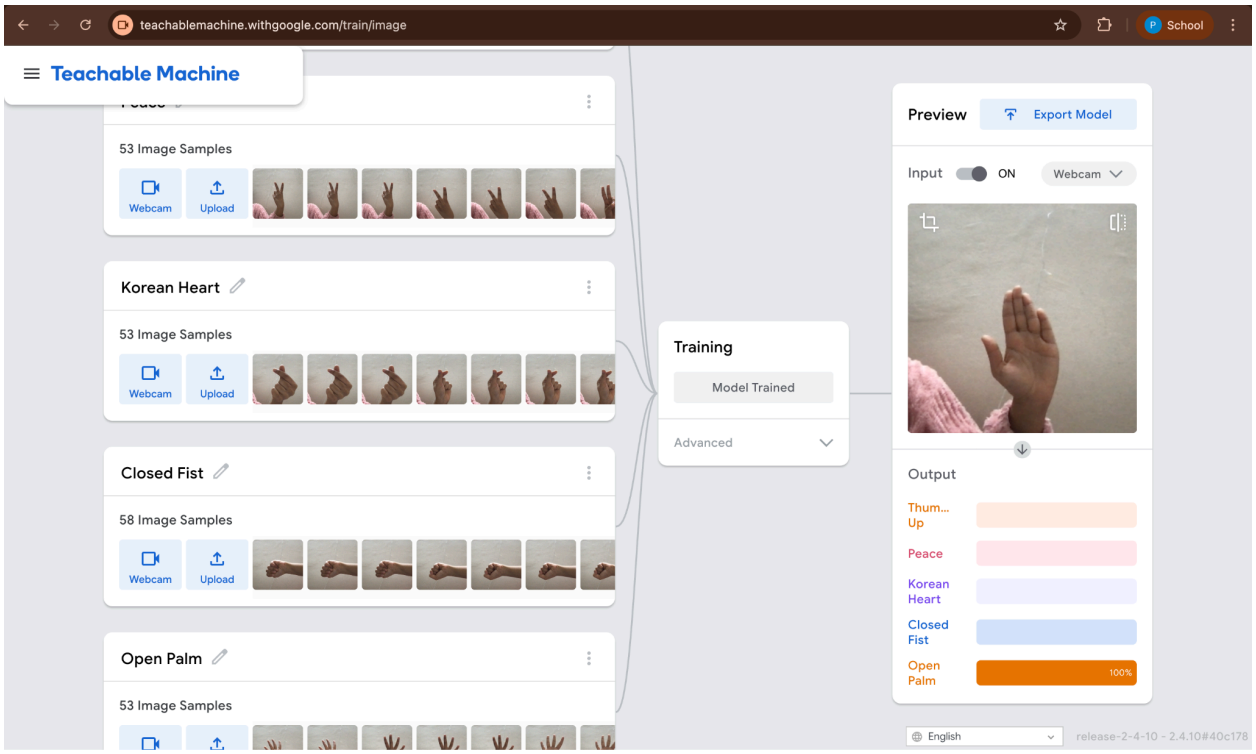
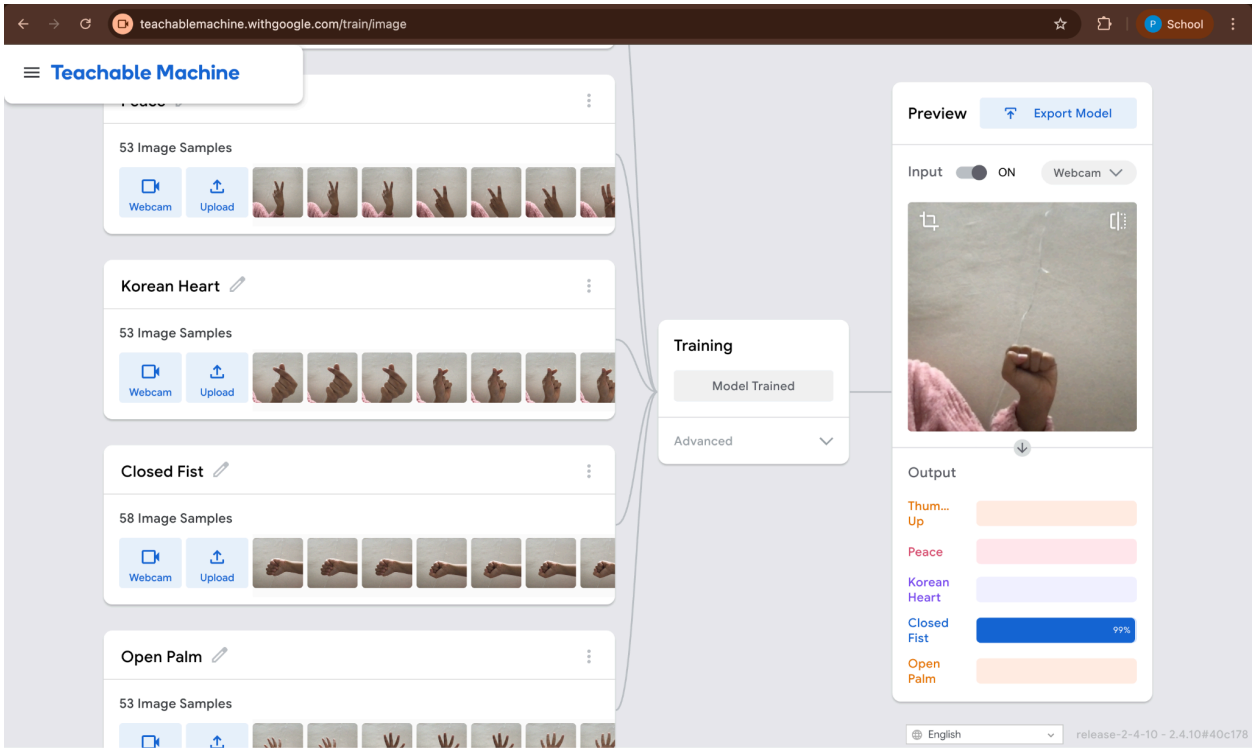
Hand Gesture Recognition Using Google Teachable Machine

Project Link:
https://teachablemachine.withgoogle.com/models/KtGrlPKhY/

Screenshots:




teachablemachine.withgoogle.com/train/image
School
Teachable Machine
53 Image Samples
Webcam
Upload
Korean Heart
53 Image Samples
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Upload
Closed Fist
58 Image Samples
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Upload
Open Palm
53 Image Samples
Training
Model Trained
Advanced
Preview
Export Model
Input
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Webcam
Output
Thum... Up
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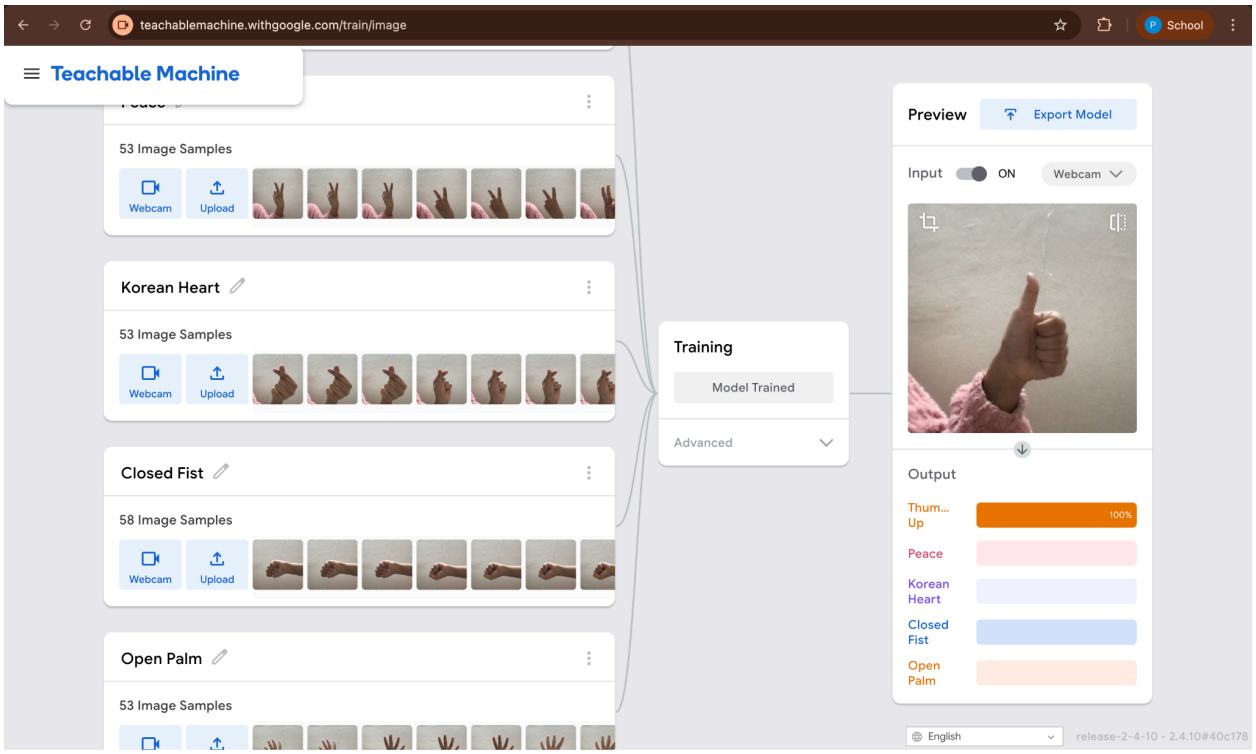
English
release-2-4-10 - 2.4.10#40c178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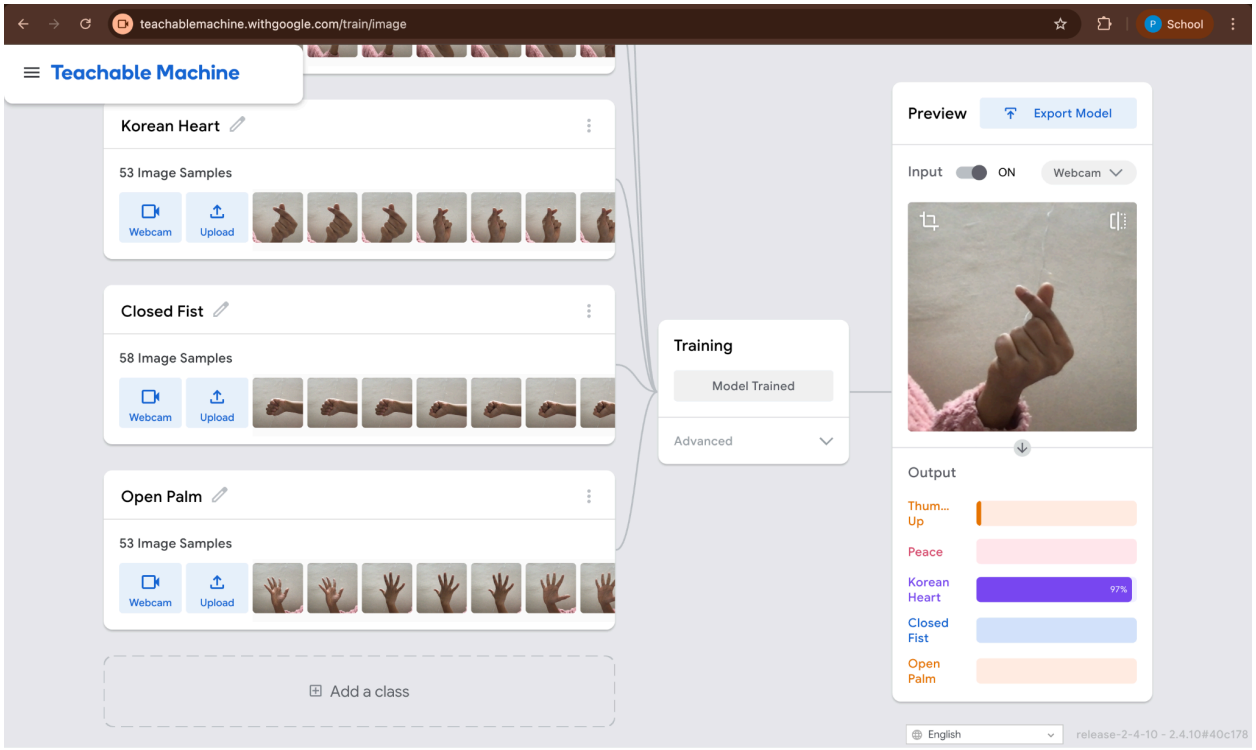
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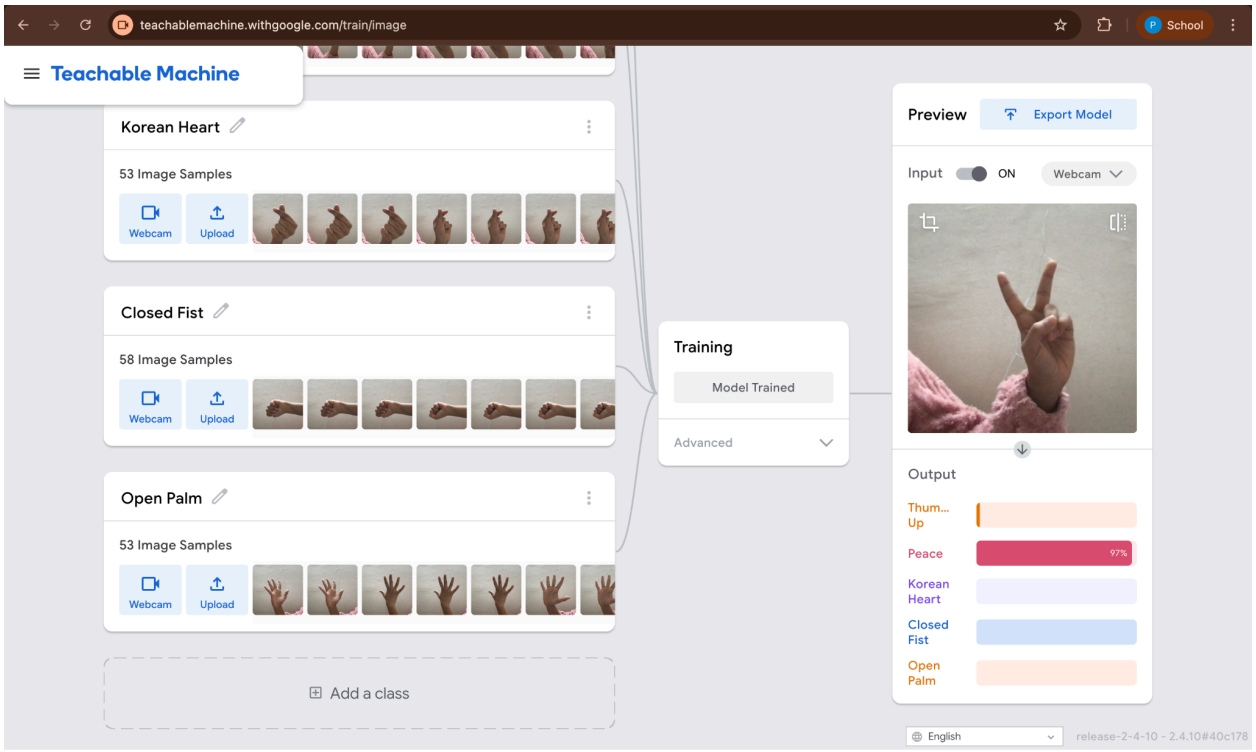
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Introduction:

For this assignment, I used Google's Teachable Machine to build a simple machine learning model that can recognise different hand gestures using a webcam. Teachable Machine is a beginner-friendly tool that allows anyone to train machine learning models without writing code. The main aim of this project was to understand how machine learning works at a basic level and to see how hand gestures can be used as an alternative way to control devices.

I trained the model on image data captured by my webcam and tested it in real time to assess how accurately it could identify different gestures. This project helped me understand how data collection, training, and testing all play an important role in building a working machine learning model.

Choice of Gestures and Reasoning:

I chose five hand gestures for this project: **Thumbs Up, Peace Sign, Korean Heart, Closed Fist, and Open Palm**. These gestures were selected because they are easy to perform and look clearly different from one another. This makes it easier for the model to learn and recognise them correctly.

Gestures like the thumbs up and open palm are very common and intuitive, making them suitable for basic command-based systems. The peace sign and closed fist were included because they involve different finger arrangements and hand shapes. I also added the Korean heart gesture to see if the model could handle more detailed finger positioning.

For each gesture, I collected at least 50 images in slightly different positions, angles, and distances from the camera. This helped make the model more flexible and less dependent on a single fixed hand pose.

Training Process and Challenges:

To train the model, I used the webcam feature in Teachable Machine and recorded images for each gesture. While recording, I tried to change my hand angle, position, and distance from the camera so the model wouldn't only learn one specific pose. This part made it clear how important data variety is in machine learning.

One of the main issues I faced was the model confusing similar gestures, especially the open palm and peace sign. When my fingers were not clearly separated, the model sometimes

predicted the wrong gesture. To fix this, I added more training images with very clear hand shapes and retrained the model.

Lighting was another challenge. When the lighting changed, the prediction confidence sometimes dropped. To improve this, I recorded additional samples under different lighting conditions so the model could better handle variation.

During testing, I noticed that some gestures worked very well while others showed lower confidence. Whenever this happened, I collected more images for that particular gesture and retrained the model. After a few rounds of training and testing, the predictions became more accurate and consistent.

This process helped me understand that training a machine learning model is not a one-time task but an iterative process that improves with better data.

Accuracy and Model Performance:

After training, the model performed quite well during live testing. Most gestures were recognised correctly with confidence levels above 90% when the hand was clearly visible. The thumbs up and closed fist gestures gave the most consistent results. The Korean heart gesture sometimes showed slightly lower confidence, but adding more training images improved its accuracy. Overall, the model reliably recognised all five gestures in real time.

Possible Real-World Applications:

This type of hand gesture recognition system can be useful in many real-world situations. It can be used for contactless control of devices, such as controlling presentations, music players, or smart home systems. It can also be helpful for people with limited mobility who may find traditional input methods difficult to use. Gesture-based systems can also be used in gaming, virtual reality, and interactive learning tools.

Conclusion:

This assignment gave me practical exposure to how machine learning models are created and improved. Using Teachable Machine made the learning process simple and interactive. Overall, the project helped me understand the importance of data quality, testing, and retraining in machine learning, and showed how gesture recognition can be applied in real life.